

## 4 Radar, electronic warfare, EMI/EMC testing

Rapid advances in radar and electronic warfare technology have created the need for leading-edge testing technology and tools. Robust radar test equipment reduce uncertainty during the design process and deliver confidence in the integrity of increasingly complex designs.

Traditional signal analysers are unable to trigger on transient problems and the maximum available acquisition bandwidth in the mid-range is just 40 MHz.

To capture transients for analysis, Tektronix RSA5000 series offers

### Key Features of Tektronix RSA5000 Series

| Features                                 | Description                                  |
|------------------------------------------|----------------------------------------------|
| Swept DPX engine                         | Collects 292,000 spectrum updates per second |
| Measurement of small spectral transients | As brief as 5.8 microseconds                 |
| Displayed average noise level            | -154 dBm/Hz @ 1 GHz                          |
| Third-order intercept                    | +17 dBm                                      |
| Discover and capture of up to 85 MHz     | Includes bands of BT, Zigbee, RFID and WLAN  |

frequency mask, frequency-edge, density, time-qualified and runt triggers. It can also be used to isolate hard-to-find hardware and software anomalies with cross domain

triggering between multiple instruments. It can capture a seamless time record of RF frequencies into deep memory for up to 7 seconds at 85MHz bandwidth.

## 5 Built for the field

For this kind of application, you require a spectrum analyser that is rugged, portable and performs well enough to get the job done. Compliance to MIL-PRF-28800 Class 2 is also important. This specification covers the general requirements for equipment used for testing and calibration of electrical and electronic equipment. The test equipment may be of commercial design and include general-purpose, special-purpose, peculiar, console-mounted, automatic test equipment and calibration standards.

The Agilent N9344C handheld spectrum analyser is built for the

### Key Features of Agilent N9344C

| Features                      | Value                              |
|-------------------------------|------------------------------------|
| Wide frequency range          | 1 MHz to 20 GHz (tunable to 9 kHz) |
| Internal preamplifier         | Up to 20 GHz                       |
| GPS                           | GPS receiver and antenna built-in  |
| Tracking generator            | Built-in                           |
| Displayed average noise level | -144 dBm                           |
| Third-order intercept         | 11 dBm @ 1 GHz                     |

field. It features a channel scanner that can measure up to 20 channels simultaneously, as well as a spectrum monitor with spectrogram display, record and playback. Moreover, it features AM/FM/ASK/FSK modulation analysis and time-gated

spectrum analysis, which allows intermittent or burst signal spectrum measurement.

The Agilent N9344C is priced at ₹ 1,329,530 for the no-frills version, while a typical configuration would cost you ₹ 1,339,269.

*The author is a tech correspondent at EBY*

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